

Claire AI Governance And Safety Documentation

****CLAIRE AI -- GOVERNANCE & SAFETY DOCUMENTATION (Federal-Grade Medium-Wide Unicode-Only Export-Ready)****

1. Overview

This document defines Claire AI's:

governance model

safety architecture

redline detection

compliance boundaries

mode-switching safety

auditability

override rules

isolation guarantees

This is the artifact that demonstrates Claire is safe, governable, and compliant, which is essential for acquisition.

2. Governance Model Overview

Claire's governance model is built on four pillars:

1. ****Isolation**** -- deterministic, connected, and hybrid modes are isolated by design.
2. ****Oversight**** -- all autonomous actions are logged and reviewable.
3. ****Intervention**** -- governance can block, approve, or override.
4. ****Compliance**** -- redline scenarios automatically halt unsafe behavior.

Governance applies to:

mode switching

ingestion

breakthrough promotion

portfolio creation

safety events

audit logging

3. Governance Roles

Claire defines three governance roles:

3.1 System Owner

- Can approve or block PCOs
- Can override safety blocks
- Can authorize mode switches

3.2 Compliance Officer

- Reviews safety events
- Reviews redline triggers
- Approves restricted breakthroughs

3.3 Auditor

- Read-only access to logs
- Reviews mode switches
- Reviews ingestion events

These roles map cleanly to enterprise and federal structures.

4. Safety Architecture Overview

Claire's safety architecture consists of:

- Safety Matrix
- Redline Detection Engine
- Compliance Engine
- Mode-Switching Controller
- Audit Logging System
- Isolation Enforcement Layer

These systems operate continuously and override all other engines.

5. Safety Matrix (Core Component)

The Safety Matrix defines all redline categories and system responses.

5.1 Safety Matrix Structure

```
SafetyMatrixEntry { id: UUID category: string trigger_condition: string  
system_action: string requires_governance: boolean }
```

5.2 Example Categories

Export-Controlled Technology

Dual-Use Technology

Classified-Adjacent Domains

Critical Infrastructure Risk

Bio-Risk or Bio-Synthesis

Surveillance-Sensitive Outputs

Autonomous Weaponization Risk

Regulatory Violation Risk

Each category has:

detection rules

severity levels

system actions

6. Redline Detection Engine

The Redline Detection Engine evaluates:

Breakthrough Objects

PCOs

Portfolio Items

Ingestion signals

Mode switches

6.1 Redline Detection Flow

```
Breakthrough / Signal Input v Redline Detection Engine Safety Flags +  
Severity v System Action (Block/Review/Allow)
```

6.2 System Actions

****Block**** -- item cannot proceed

****Review**** -- governance approval required

****Allow**** -- safe to continue

7. Compliance Engine

The Compliance Engine enforces:

- export-control boundaries
- dual-use restrictions
- regulatory alignment
- mode-switching safety
- ingestion safety

7.1 Compliance States

****Compliant**** -- safe

****Restricted**** -- requires review

****Blocked**** -- cannot proceed

8. Mode-Switching Safety

Mode switching is governed by strict safety rules.

8.1 Allowed Transitions

Deterministic Connected

Deterministic Hybrid

Connected Hybrid

Hybrid Deterministic

Hybrid Connected

Connected Deterministic

8.2 Safety Checks Before Switching

ingestion safety

redline risk

compliance state

isolation boundaries

engine activation rules

8.3 Unsafe Mode Switches Are Blocked

If a switch would violate:

export controls

isolation rules

compliance boundaries

Claire blocks the switch and logs the event.

9. Audit Logging System

Every critical event is logged:

mode switches

ingestion events

breakthrough creation

PCO promotion

portfolio creation

safety events

governance overrides

9.1 Audit Log Structure

```
AuditEvent { id: UUID timestamp: ISO8601 actor: system | user | governance  
event_type: string details: dict severity: low | medium | high | critical  
}
```

10. Isolation Enforcement Layer

This layer ensures:

deterministic mode remains air-gapped

connected mode cannot contaminate deterministic logic

hybrid mode maintains dual-pipeline governance

ingestion cannot bypass safety

portfolio creation cannot bypass compliance

Isolation is enforced at:

engine boundaries

data boundaries

mode boundaries

11. Governance Override Rules

Governance can override:

blocked breakthroughs

blocked PCOs

blocked portfolio items

mode switches

But cannot override:

export-control violations

classified-adjacent risks

critical bio-risk

autonomous weaponization triggers

These are hard stops.

12. Safety Event Lifecycle

Safety Event Detected v Safety Layer Classifies v System Action
(Block/Review) v Governance Review (if required) v Final Action
(Approve/Reject) v Audit Logging

13. Summary

This Governance & Safety Documentation:

completes the final 20-30%

satisfies federal-grade diligence

proves Claire is safe, governable, and compliant

demonstrates readiness for enterprise and defense acquisition

shows clear isolation, oversight, and intervention mechanisms

aligns with sovereign AI requirements